Functional Specification Document (FSD)

Order-to-Cash (O2C) & Inventory Automation System | Client (fictional): PT Maju Distribusi Nusantara | Prepared by: Alief Sukma Dewanta, Business Analyst / System Analyst | Version 1.0

> *Portfolio note: the client is fictional and all figures are simulated. This FSD specifies how the system behaves to meet the business requirements (BRQ) in the companion Business Requirements Document (BRD). Requirement IDs (FR, BR, NFR) are the same IDs used in the traceability matrix and test workbook.*

1. Document Control

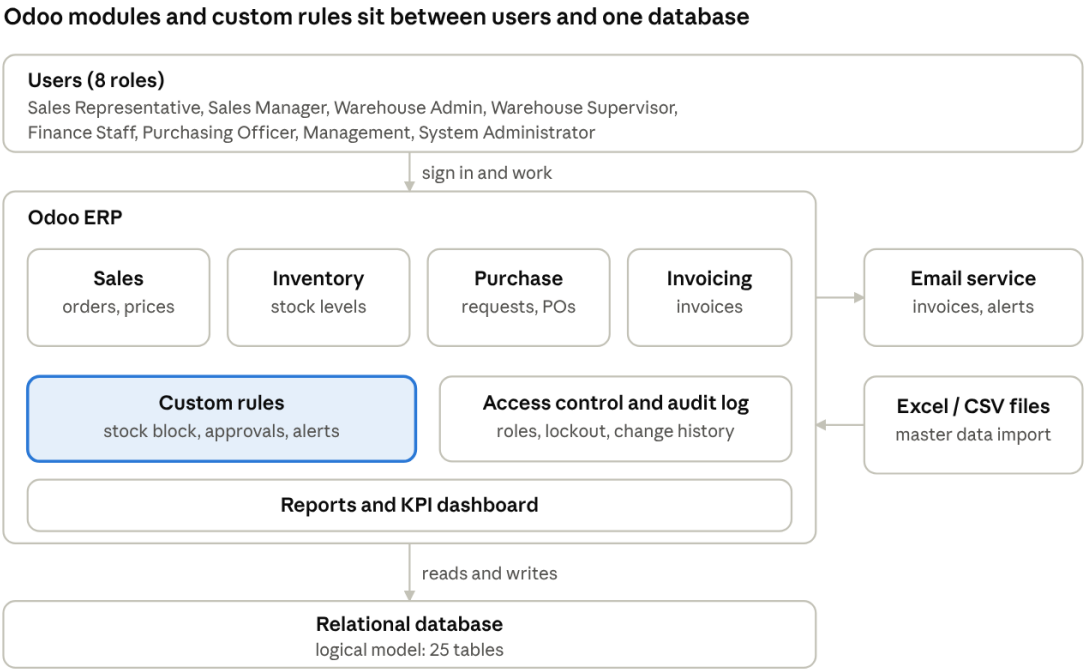
Item	Detail
Document	Functional Specification Document
Companion document	Business Requirements Document (BRD)
Status	Draft for review
Target platform	Odoo ERP (Sales, Inventory, Purchase, Invoicing) plus configured rules
Related files	Requirements workbook, test management workbook (RTM, test cases, UAT), wireframes

Version	Date	Author	Change
0.1	Nov 2026	A. S. Dewanta	Combined BRD/FSD draft
1.0	Nov 2026	A. S. Dewanta	Split into FSD; roles, screens, validation, and diagrams added

2. Purpose and Scope

This document defines the functional behaviour, business rules enforcement, data model, screens, notifications, and non-functional requirements of the O2C system. It covers 46 functional requirements in seven modules and 7 non-functional requirements. Scope boundaries (in and out of scope) are defined in the BRD, section 5. Payment gateway integration, transport routing, and multi-warehouse support are out of scope.

3. Solution Overview



solution architecture · users, Odoo modules, custom rules, database

Component	Responsibility
Sales	Customers, price lists, quotations and orders, approvals
Inventory	On hand, reserved and available quantities, reservations, delivery orders, backorders, adjustments, counts
Purchase	Reordering rules, purchase requests, purchase orders, goods receipts
Invoicing	Invoices from delivered quantities, payment terms, payments, credit notes, receivables aging
Custom rules	Overselling block, approval routing, notifications that standard configuration does not provide
Access control and audit	Roles, permissions, account lockout, record of changes
Reports and dashboard	Stock report, sales report, KPI dashboard, Excel and CSV export

The platform fit for each requirement area (Fit, Config, Custom) is recorded in the Odoo fit-gap analysis and must be verified in the installed version.

4. Users and Roles

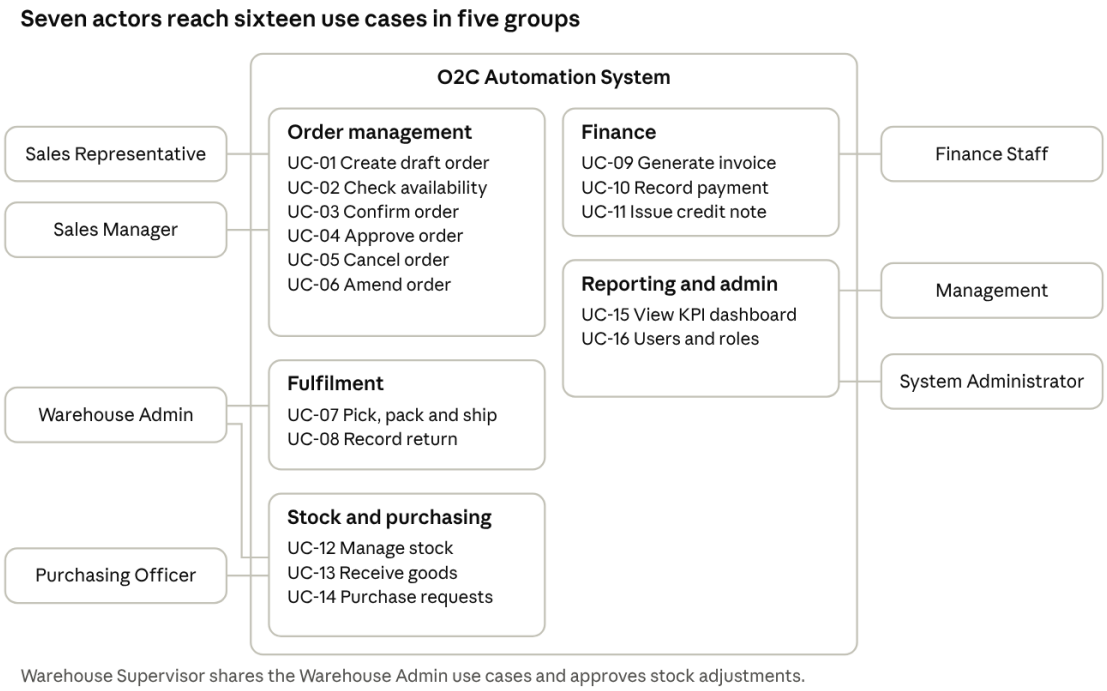
Role	Description
Sales Representative	Creates and confirms orders for assigned customers
Sales Manager	Approves large orders, high discounts, and credit exceptions; sees all orders
Warehouse Admin	Picks, ships, records returns and goods receipts, requests stock adjustments
Warehouse Supervisor	Approves stock adjustments and over-receipts; manages cycle counts
Finance Staff	Manages invoices, payments, and credit notes
Purchasing Officer	Handles purchase requests and supplier orders
Management	Read-only access to reports and the KPI dashboard (Operations Director)
System Administrator	Manages users, roles, and configuration

4.1 Permission Matrix

Function	Sales Rep	Sales Mgr	WH Admin	WH Supervisor	Finance Staff	Purchasing	Management	System Admin
Create and edit draft order	Own customers	All	-	-	-	-	-	-
Confirm order	Own customers	All	-	-	-	-	-	-
Approve order	-	Yes	-	-	-	-	-	-
Cancel order before shipment	Own	All	-	-	-	-	-	-
View stock levels	View	View	View	Full	View	View	View	-
Request stock	-	-	Yes	Yes	-	-	-	-

Function	Sales Rep	Sales Mgr	WH Admin	WH Supervisor	Finance Staff	Purchasing	Management	System Admin
adjustment								
Approve stock adjustment	-	-	-	Yes	-	-	-	-
Pick and ship	-	-	Yes	Yes	-	-	-	-
Record return	-	-	Yes	Yes	View	-	-	-
Goods receipt	-	-	Yes	Yes	-	View	-	-
View invoices	Own orders	View	-	-	Yes	-	View	-
Record payment and credit note	-	-	-	-	Yes	-	-	-
Purchase requests and orders	-	-	-	View	-	Yes	View	-
KPI dashboard	Own	Yes	-	Yes	Yes	-	Yes	-
View audit log	-	View	-	-	-	-	View	Yes
Manage users and roles	-	-	-	-	-	-	-	Yes

5. Use Cases



use case overview · 7 actors, 16 use cases in 5 groups

ID	Use case	Primary actor	Trigger	Requirements
UC-01	Create draft order	Sales Representative	Customer order received	FR-ORD-01, FR-ORD-04, FR-MD-01, FR-MD-02
UC-02	Check availability	Sales Representative	Adding a product line	FR-ORD-02, FR-INV-01
UC-03	Confirm order	Sales Representative	Order ready	FR-ORD-03, FR-ORD-07, FR-INV-02
UC-04	Approve order	Sales Manager	Order in Pending Approval	FR-ORD-05, FR-ORD-06
UC-05	Cancel order	Sales Representative	Customer cancels	FR-ORD-08, FR-INV-03
UC-06	Amend confirmed order	Sales Representative	Customer changes quantity	FR-ORD-09
UC-07	Pick, pack, and ship	Warehouse Admin	Order confirmed	FR-SHP-01 to FR-SHP-05, FR-INV-04
UC-08	Record return	Warehouse Admin	Goods returned	FR-SHP-06

ID	Use case	Primary actor	Trigger	Requirements
UC-09	Generate invoice	System	Shipment marked Shipped	FR-FIN-01 to FR-FIN-03
UC-10	Record payment	Finance Staff	Payment received	FR-FIN-04, FR-FIN-07
UC-11	Issue credit note	Finance Staff	Approved return or correction	FR-FIN-05, FR-FIN-06
UC-12	Manage stock	Warehouse Admin, Supervisor	Variance found, count scheduled	FR-INV-05, FR-INV-08, FR-INV-09
UC-13	Receive goods	Warehouse Admin	Supplier delivery	FR-INV-07
UC-14	Manage purchase requests	Purchasing Officer	Low-stock alert	FR-INV-06
UC-15	View KPI dashboard	Management	Periodic review	FR-RPT-01 to FR-RPT-04
UC-16	Manage users and roles	System Administrator	Staff change	FR-SEC-01 to FR-SEC-04

6. Functional Requirements

Priority uses MoSCoW: **M** = Must, **S** = Should, **C** = Could. Summary: 46 requirements (31 Must, 13 Should, 2 Could).

6.1 Master Data (MD)

ID	Requirement	Priority
FR-MD-01	The system shall allow creating and editing customer records with name, address, contact, payment terms, credit limit, and assigned sales rep.	M
FR-MD-02	The system shall allow creating and editing product records with SKU, name, unit of measure, price, and reorder point.	M
FR-MD-03	The system shall support customer-specific price lists that override the default product price.	S
FR-MD-04	The system shall prevent duplicate SKUs and duplicate customer tax IDs.	M
FR-MD-05	The system shall support bulk import of customers and products from CSV or Excel with row-level error reporting.	S

6.2 Sales and Order Management (ORD)

ID	Requirement	Priority
FR-ORD-01	The system shall allow a Sales Rep to create a draft order for a customer with one or more product lines.	M
FR-ORD-02	The system shall display real-time available stock (on hand minus reserved) for each product while the order is being entered.	M
FR-ORD-03	The system shall block order confirmation when any line quantity exceeds available stock and show the available quantity (BR-01).	M
FR-ORD-04	The system shall calculate line totals, discounts, tax, and order total automatically.	M
FR-ORD-05	The system shall route orders above IDR 50,000,000 or discounts above 10% to the Sales Manager for approval (BR-04, BR-09).	M
FR-ORD-06	The system shall check the customer's credit limit and overdue balance at confirmation and block or route for approval (BR-03).	S
FR-ORD-07	The system shall move orders through the statuses Draft, Pending Approval, Confirmed, Processing, Partially Shipped, Shipped, Completed, and Cancelled.	M
FR-ORD-08	The system shall allow cancelling an order before shipment and release its reserved stock (BR-06).	M
FR-ORD-09	The system shall allow editing a confirmed order only through an amendment that re-validates stock and is logged.	S
FR-ORD-10	The system shall send an order confirmation email or document to the customer.	C

6.3 Inventory Management (INV)

ID	Requirement	Priority
FR-INV-01	The system shall maintain On Hand, Reserved, and Available quantities per product, where Available = On Hand minus Reserved.	M
FR-INV-02	The system shall reserve stock automatically when an order is confirmed (BR-02).	M
FR-INV-03	The system shall release reserved stock when an order is cancelled or an amendment reduces quantity.	M
FR-INV-04	The system shall reduce On Hand quantity when a shipment is confirmed.	M

ID	Requirement	Priority
FR-INV-05	The system shall support manual stock adjustments with a mandatory reason code and Supervisor approval.	M
FR-INV-06	The system shall generate a low-stock alert and draft purchase request when Available falls below the reorder point (BR-07).	M
FR-INV-07	The system shall support goods receipt against a purchase order and increase On Hand accordingly.	M
FR-INV-08	The system shall support cycle count sessions and show variance between system and counted quantity.	S
FR-INV-09	The system shall show a stock movement history per product.	S

6.4 Warehouse Fulfilment and Shipping (SHP)

ID	Requirement	Priority
FR-SHP-01	The system shall notify the Warehouse team and create a pick list when an order is confirmed.	M
FR-SHP-02	The system shall allow the Warehouse Admin to record picked quantities per line.	M
FR-SHP-03	The system shall allow partial shipment when picked quantity is less than ordered, keeping the remainder as a backorder.	S
FR-SHP-04	The system shall generate a delivery note for each shipment.	M
FR-SHP-05	The system shall allow marking a shipment as Shipped and later as Delivered with a timestamp and receiver name.	M
FR-SHP-06	The system shall support recording a customer return with reason and restocking decision.	S

6.5 Finance and Invoicing (FIN)

ID	Requirement	Priority
FR-FIN-01	The system shall automatically generate an invoice when a shipment is marked Shipped (BR-05).	M
FR-FIN-02	The system shall produce a PDF invoice and email it to the customer's billing contact.	M
FR-FIN-03	The system shall set the invoice due date from the customer's payment terms.	M

ID	Requirement	Priority
FR-FIN-04	The system shall allow recording full or partial payments against an invoice and update payment status (Unpaid, Partially Paid, Paid, Overdue).	M
FR-FIN-05	The system shall lock a sent invoice and allow corrections only via credit note (BR-10).	M
FR-FIN-06	The system shall create a credit note linked to a return or an invoice correction.	S
FR-FIN-07	The system shall show an accounts receivable aging report (current, 1-7, 8-30, over 30 days).	S
FR-FIN-08	The system shall send an automatic payment reminder for invoices past due.	C

6.6 Reporting and Dashboard (RPT)

ID	Requirement	Priority
FR-RPT-01	The system shall provide a dashboard showing order processing time, order volume by status, and overselling blocks.	S
FR-RPT-02	The system shall provide a stock level report with low-stock and stockout indicators.	M
FR-RPT-03	The system shall provide a sales report by customer, product, and sales rep for a selected period.	S
FR-RPT-04	The system shall allow exporting reports to Excel or CSV.	S

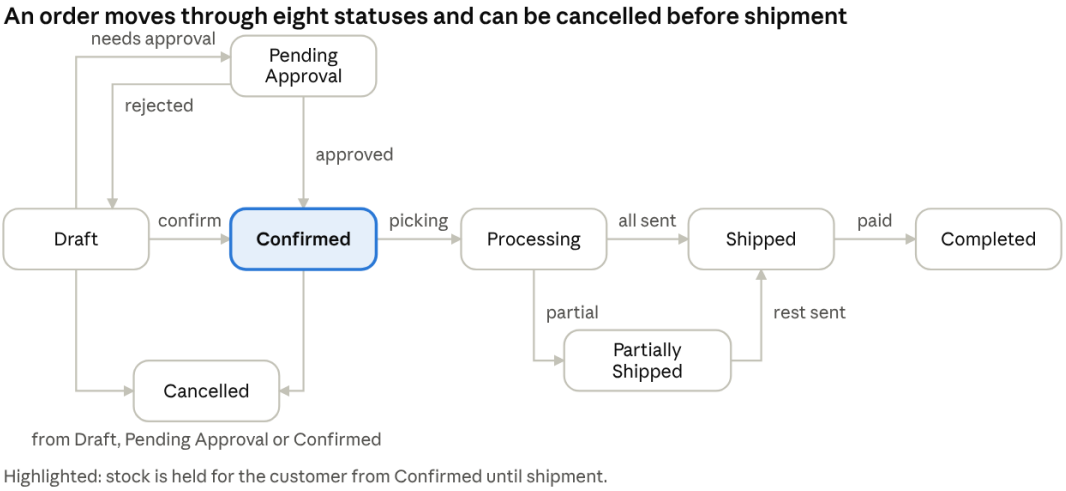
6.7 Security and Audit (SEC)

ID	Requirement	Priority
FR-SEC-01	The system shall enforce role-based access control as defined in section 4.	M
FR-SEC-02	The system shall record an audit log of status changes, price overrides, approvals, and stock adjustments with user and timestamp (BR-08).	M
FR-SEC-03	The system shall require authentication and lock an account after 5 failed login attempts.	M
FR-SEC-04	The system shall allow the System Administrator to create, disable, and reassign users and roles.	M

7. Business Rules and Where the System Enforces Them

Rule	Statement	Enforced at	Related FR
BR-01	Order cannot be confirmed if quantity exceeds available stock	Order confirmation, atomic stock update	FR-ORD-03
BR-02	Confirmation reserves stock; reserved stock is not available to others	Order confirmation	FR-INV-02
BR-03	Credit limit and overdue balance control for credit orders	Order confirmation (customer check)	FR-ORD-06
BR-04	Orders above IDR 50,000,000 need Sales Manager approval	Order confirmation (total check)	FR-ORD-05
BR-05	Invoice created only when goods (or part) are shipped	Shipment confirmation	FR-FIN-01
BR-06	Cancel releases stock; no cancellation after shipment	Cancel action, status check	FR-ORD-08, FR-INV-03
BR-07	Low available stock creates alert and draft purchase request	After every stock change	FR-INV-06
BR-08	Critical changes are logged with user and time	All status, price, approval, adjustment actions	FR-SEC-02
BR-09	Discounts above 10% need Sales Manager approval	Order confirmation (discount check)	FR-ORD-05
BR-10	Sent invoices are locked; corrections use credit notes	Invoice edit action	FR-FIN-05

8. Order Status Model

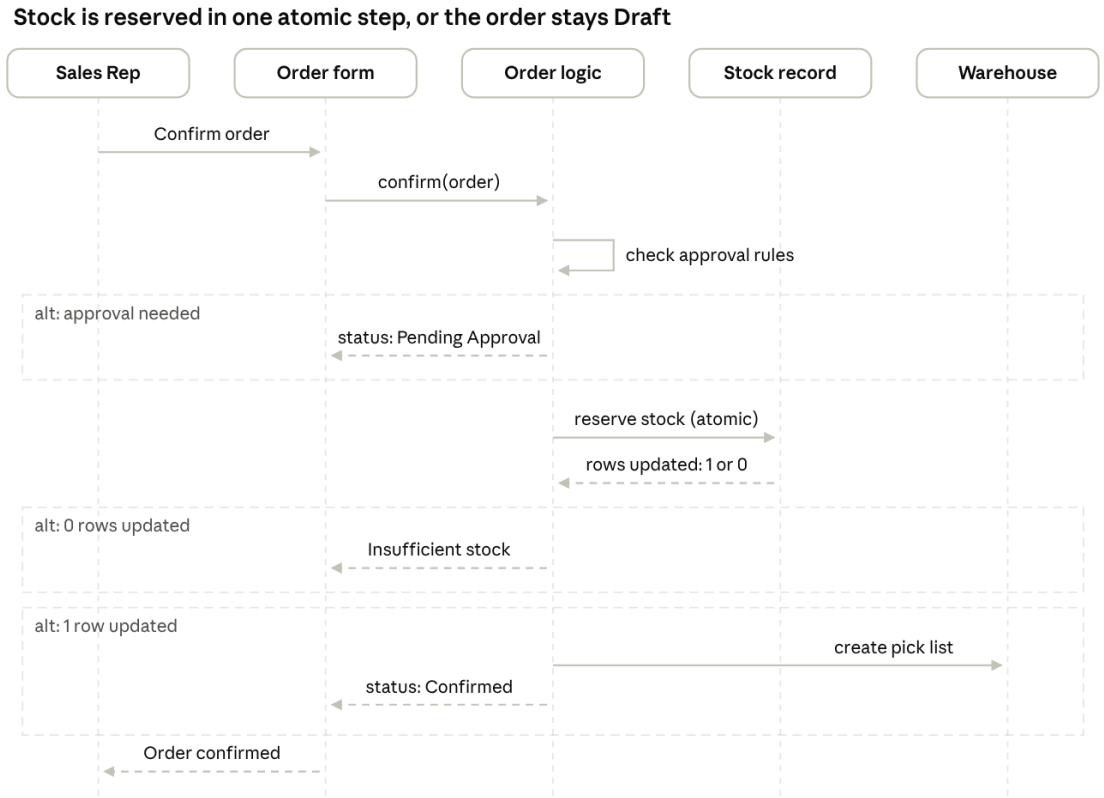


order status diagram · 8 statuses, 11 transitions

From	To	Trigger	Rule
Draft	Pending Approval	Confirm with approval condition met	BR-03, BR-04, BR-09
Draft	Confirmed	Confirm, stock sufficient	BR-01, BR-02
Pending Approval	Confirmed	Manager approves	Stock re-validated
Pending Approval	Draft	Manager rejects	Reason required
Confirmed	Processing	Warehouse starts picking	
Processing	Partially Shipped	Partial shipment confirmed	FR-SHP-03
Processing or Partially Shipped	Shipped	Full shipment confirmed	Invoice generated, BR-05
Shipped	Completed	Delivery confirmed and invoice fully paid	
Draft, Pending Approval, or Confirmed	Cancelled	Cancel before shipment	BR-06

9. Process Logic

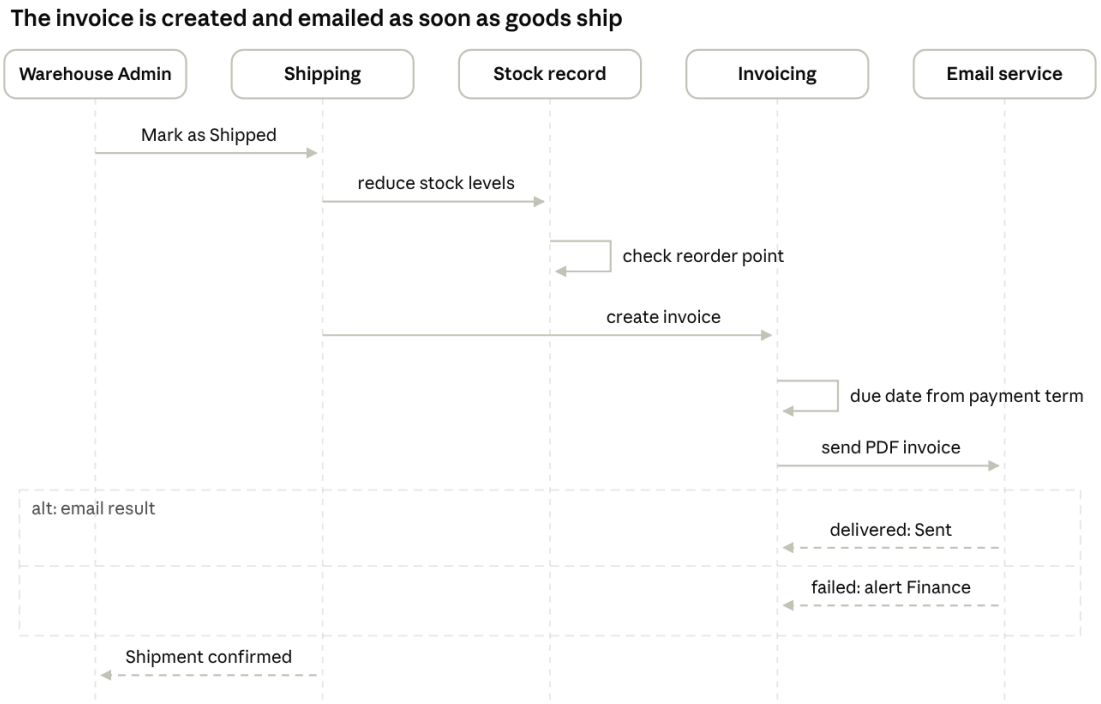
9.1 Confirming an Order and Reserving Stock



sequence diagram · confirm order and reserve stock

1. The Sales Rep confirms the order. The system checks the approval rules (credit limit, order total, discount).
2. If an approval is needed, the status becomes Pending Approval and no stock is reserved.
3. Otherwise the system reserves stock with one atomic update that succeeds only if available stock covers the quantity and the stock record has not changed since it was read.
4. If no row is updated, the order stays Draft and the user sees "Insufficient stock. Available: X". This is also how simultaneous orders for the last units are handled (NFR-02): one succeeds and the other is rejected.
5. On success the order becomes Confirmed, a pick list is created for the warehouse, and the change is written to the audit log.

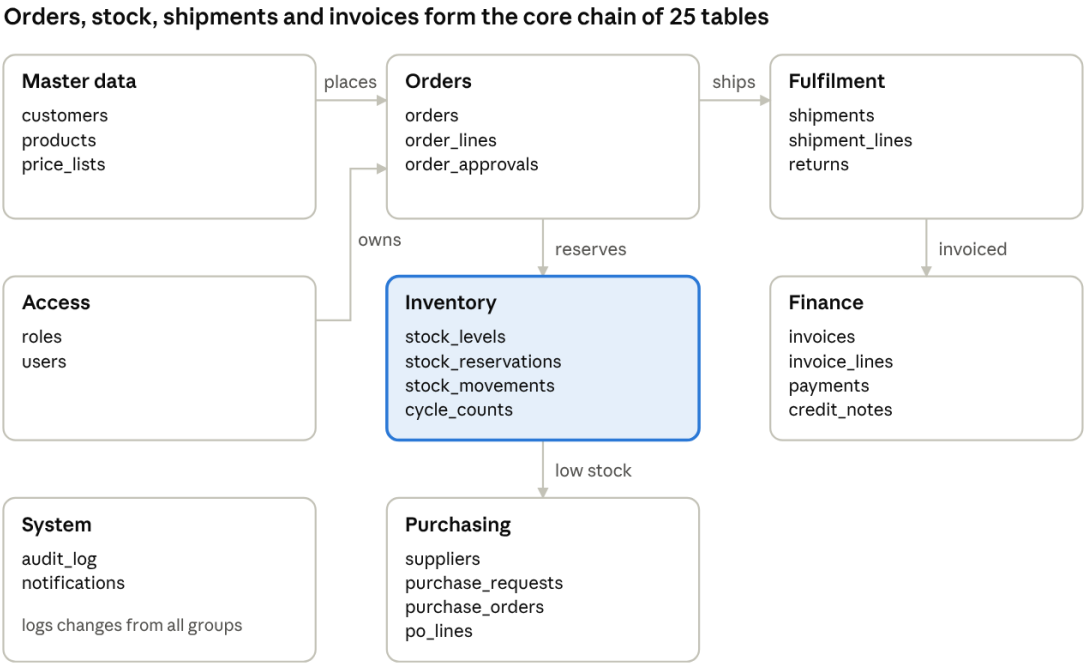
9.2 Shipping and Invoicing



sequence diagram · ship, invoice, and email

1. The Warehouse Admin marks the shipment as Shipped (full or partial).
2. The system reduces On Hand and Reserved by the shipped quantity and checks the reorder point.
3. The system creates an invoice for the shipped lines only, sets the due date from the payment term, and emails the PDF.
4. If the email fails, the invoice is saved with status Failed and Finance is alerted.

10. Data Model



data model overview · 25 tables in 8 groups

The logical model has 25 tables. Available stock is derived (On Hand minus Reserved) and never stored. The full column list, keys, and constraints are in the ERD documentation and DBML file.

Group	Tables
Access	roles, users
Master data	customers, products, price_lists
Orders	orders, order_lines, order_approvals
Inventory	stock_levels, stock_reservations, stock_movements, cycle_counts
Fulfilment	shipments, shipment_lines, returns
Finance	invoices, invoice_lines, payments, credit_notes
Purchasing	suppliers, purchase_requests, purchase_orders, po_lines
System	audit_log, notifications

Key design decisions

1. On Hand and Reserved quantities are stored separately; Available is calculated.
2. Reservations are kept per order line so cancellations and partial shipments release or consume exactly the right units.

3. Stock records carry a version number so that concurrent confirmations cannot both reserve the last units.
4. Invoices link to shipments, so partial shipments produce partial invoices.
5. Sent invoices are locked; corrections use credit notes.
6. The audit log is insert-only.

11. Screens and Navigation

Wireframes for five of these screens are available as an editable canvas (order entry, stock levels, pick list and shipment, invoices, KPI dashboard).

ID	Screen	Main fields and actions	Roles	Requirements
SCR-01	Login	Email, password; lockout message	All	FR-SEC-03
SCR-02	Customers	List, create, edit, import CSV	Sales Mgr, Admin	FR-MD-01, FR-MD-04, FR-MD-05
SCR-03	Products and price lists	List, create, edit, price list	Sales Mgr, Admin	FR-MD-02, FR-MD-03
SCR-04	Sales order (wireframe 1)	Customer, lines, available stock, totals, Save draft, Confirm	Sales Rep, Sales Mgr	FR-ORD-01 to FR-ORD-04, FR-ORD-08
SCR-05	Approval queue	Orders pending approval, approve, reject with reason	Sales Mgr	FR-ORD-05, FR-ORD-06
SCR-06	Stock levels (wireframe 2)	On Hand, Reserved, Available, reorder point, status; adjustment, count, history	Warehouse, Purchasing	FR-INV-01, FR-INV-05, FR-INV-08, FR-INV-09, FR-RPT-02
SCR-07	Pick list and shipment (wireframe 3)	Ordered, picked, backorder; delivery note; Mark shipped	Warehouse	FR-SHP-01 to FR-SHP-05
SCR-08	Returns	Return lines, reason, restock choice	Warehouse	FR-SHP-06
SCR-09	Invoices and payments (wireframe 4)	Aging tiles, status, email status, record payment, credit note	Finance	FR-FIN-01 to FR-FIN-07
SCR-10	Purchase requests and orders	Draft requests, convert to PO, goods receipt	Purchasing, Warehouse	FR-INV-06, FR-INV-07
SCR-11	KPI dashboard (wireframe 5)	KPI tiles, charts, date and sales rep filters, export	Management, managers	FR-RPT-01, FR-RPT-03, FR-RPT-04

ID	Screen	Main fields and actions	Roles	Requirements
SCR-12	Users and roles	Create, disable, assign roles; audit log view	Admin	FR-SEC-01, FR-SEC-02, FR-SEC-04

12. Validation Rules and Messages

ID	Rule	Message shown
VAL-01	Line quantity must be greater than 0	Quantity must be greater than 0
VAL-02	Quantity must not exceed available stock at confirmation	Insufficient stock. Available: [X]
VAL-03	Discount must be between 0 and 100%	Discount must be between 0 and 100
VAL-04	SKU and customer tax ID must be unique	A record with this [SKU / tax ID] already exists
VAL-05	A rejection needs a reason	Please enter a reason for the rejection
VAL-06	Shipped orders cannot be cancelled	Use a return for shipped orders
VAL-07	Stock adjustment needs a reason code	Reason code is required
VAL-08	An adjustment cannot make stock negative	Adjustment would result in negative stock
VAL-09	Receipt above the ordered quantity needs approval	Over-receipt needs Supervisor approval
VAL-10	Payment must not exceed the outstanding balance	Payment exceeds outstanding balance
VAL-11	A sent invoice cannot be edited	This invoice is locked. Create a credit note to correct it
VAL-12	Status changes must follow the status model	This status change is not allowed
VAL-13	Account locks after 5 failed logins	Account locked. Contact your administrator

13. Notifications and Documents

Event	Recipient	Channel	Requirement
Order needs approval	Sales Manager	In-system notification	FR-ORD-05
Order confirmed	Warehouse team	Pick list and notification	FR-SHP-01
Shipment marked Shipped	Customer billing	Email with PDF invoice	FR-FIN-02

Event	Recipient	Channel	Requirement
	contact		
Invoice email failed	Finance Staff	In-system alert	FR-FIN-02
Available stock below reorder point	Purchasing Officer	In-system notification and draft request	FR-INV-06
Over-receipt or adjustment pending	Warehouse Supervisor	In-system notification	FR-INV-05
Invoice overdue	Customer, Finance	Email reminder (Could)	FR-FIN-08
Account locked	System Administrator	In-system notification	FR-SEC-03

Documents produced: order confirmation (Could), delivery note, PDF invoice, credit note.

14. Reports and KPI Definitions

KPI	Definition	Source data
BO-01 Order processing time	Average of (pick started minus order confirmed) in hours	Order timestamps
BO-02 Overselling rate	Orders confirmed without enough stock divided by all orders	Order and reservation log
BO-03 Stock accuracy	Counted items matching system divided by counted items	Cycle counts
BO-04 Invoice lead time	Average of (invoice sent minus shipped) in minutes	Shipments, invoices
BO-05 Order error rate	Orders corrected manually divided by all orders	Audit log amendments
BO-06 Stockouts per month	Days with Available = 0, counted per product and month	Stock movements

Reports: stock level report, sales report by customer, product, and sales rep, receivables aging (current, 1-7, 8-30, over 30 days), KPI dashboard. All reports can be exported to Excel or CSV.

15. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Stock validation shall respond within 2 seconds for 95% of requests.
NFR-02	Concurrency	When two users confirm orders for the last units at the same time, only one shall succeed and the other shall receive an

ID	Category	Requirement
		insufficient-stock message.
NFR-03	Availability	The system shall be available at least 99% during working hours (07:00 to 18:00 WIB).
NFR-04	Security	Passwords shall be stored hashed; all traffic shall use HTTPS.
NFR-05	Auditability	Audit log entries shall be immutable and retained for at least 3 years.
NFR-06	Usability	A new Sales Rep shall be able to enter an order after a 30-minute training session.
NFR-07	Backup	Daily database backup with a recovery point of 24 hours.

16. Data Migration and Integration

Item	Approach
Customers and products	Clean the Excel files, import by CSV, validate duplicates (FR-MD-04, FR-MD-05)
Opening stock	Load through an inventory adjustment with a reason code and a physical count
Price lists	Import customer-specific prices where they exist
Email	Outgoing SMTP service for invoices and alerts; test with a test mailbox before go-live
WhatsApp	No integration in phase 1; orders are entered by sales reps (BRD assumption A-01)

17. Traceability

Every functional requirement traces to a user story or backlog item and at least one test case in the requirements workbook and the test management workbook (46 of 46 requirements have a test case). Business requirements in the BRD (BRQ-01 to BRQ-13) trace to the FR groups listed there.

18. Assumptions and Open Issues

ID	Item	Owner	Status
OI-01	Confirm whether the chosen Odoo edition supports order approval thresholds or needs a workaround	Business Analyst	Open
OI-02	Confirm that standard Odoo blocks or only warns on insufficient stock; implement the block as a rule if it	Business Analyst	Open

ID	Item	Owner	Status
	only warns		
OI-03	Confirm login lockout behaviour in the installed version	System Administrator	Open
OI-04	Decide reorder quantity formula (fixed quantity or maximum minus available)	Purchasing Officer	Open
OI-05	Confirm invoice email template and sender address	Finance Manager	Open

Appendix A: Additional Design Diagrams

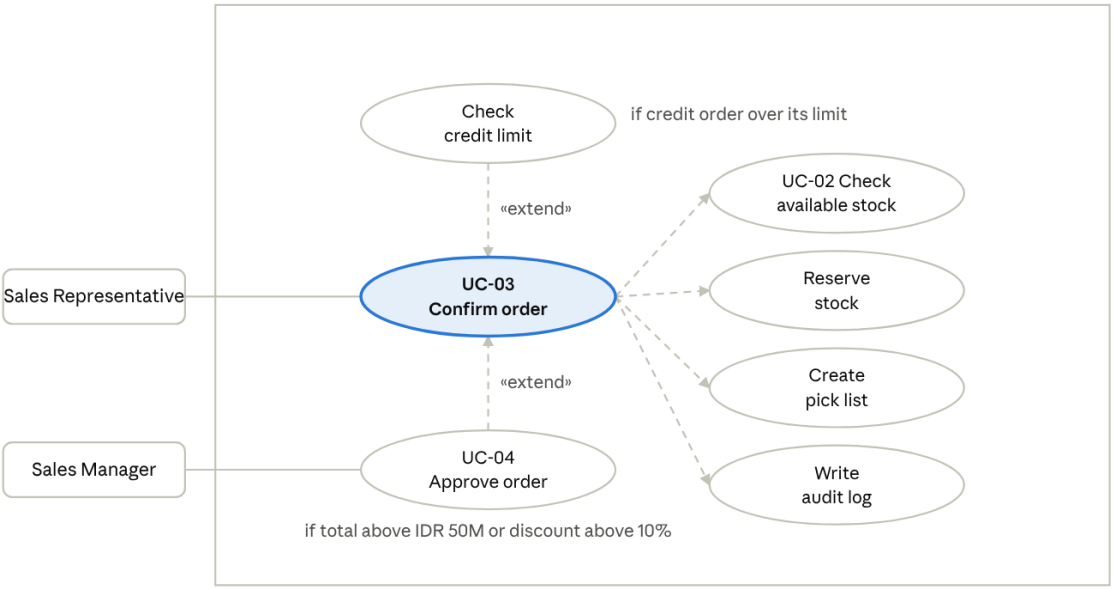
These diagrams add detail to sections 3, 5, 8, 9, and 10. The table shows where every diagram type expected in a functional specification appears.

Diagram type	Where
System architecture	Section 3
Use case (overview)	Section 5
Use case (detailed, include and extend)	A.1
User flow	A.2
Activity diagram (detailed logic)	A.3
Sequence diagrams	Sections 9.1 and 9.2
State machine (order status)	Section 8
Component diagram	A.4
Class diagram	A.5
Entity relationship diagram	Section 10 and the ERD file
Integration (API) flowchart	A.6

A.1 Detailed Use Case: Confirm Order

Confirm order always checks available stock, reserves stock, creates the pick list, and writes the audit log (include). Two steps run only when their condition is true (extend): the credit check for credit orders, and manager approval for large orders or high discounts. The pick list goes to the warehouse team.

Confirm order includes four steps and is extended by two conditional ones



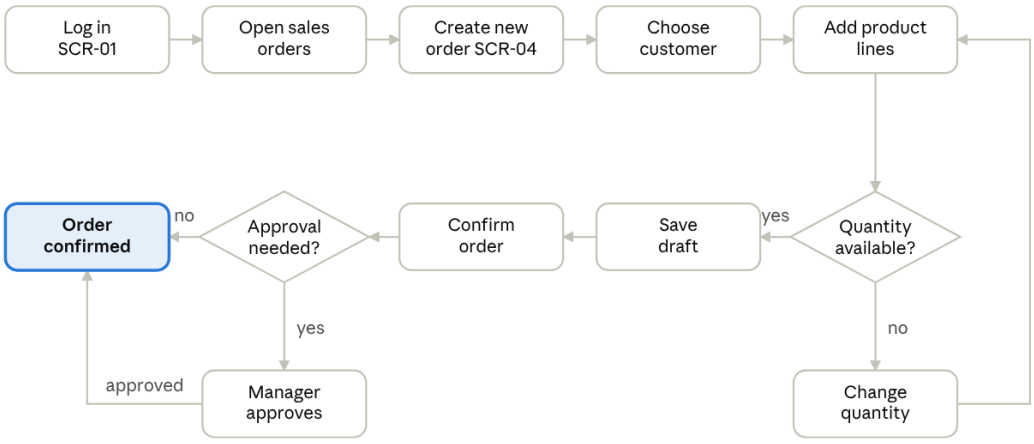
Dashed arrow to the right: always part of Confirm order. Dashed arrow from above or below: only when its condition is true.

detailed use case · Confirm order with include and extend

A.2 User Flow: From Login to a Confirmed Order

This is the path a Sales Representative follows on the screens SCR-01 and SCR-04. If the quantity is not available the user changes it and tries again; if approval is needed the order waits for the Sales Manager.

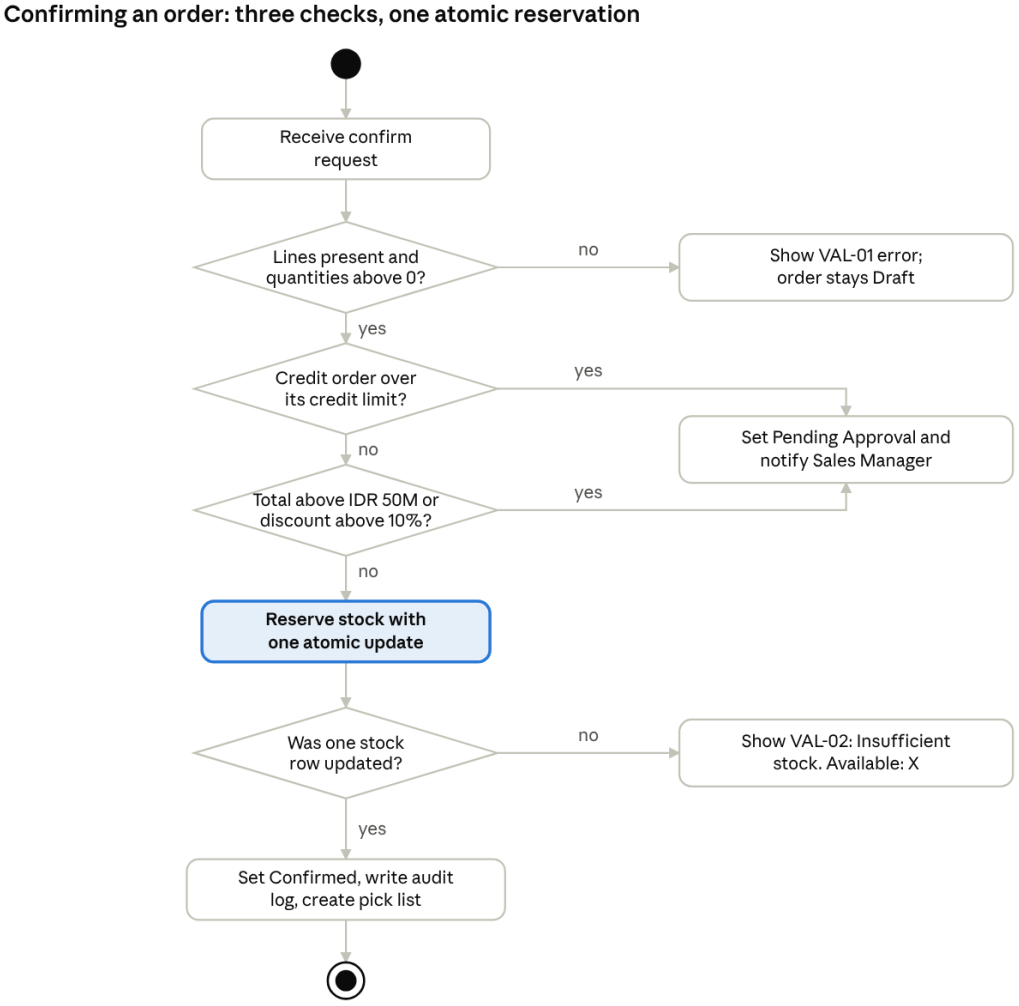
The sales rep path from login to a confirmed order



user flow · login to confirmed order

A.3 Activity Diagram: Order Confirmation Checks

The checks run in this order. Messages VAL-01 and VAL-02 are defined in section 12.

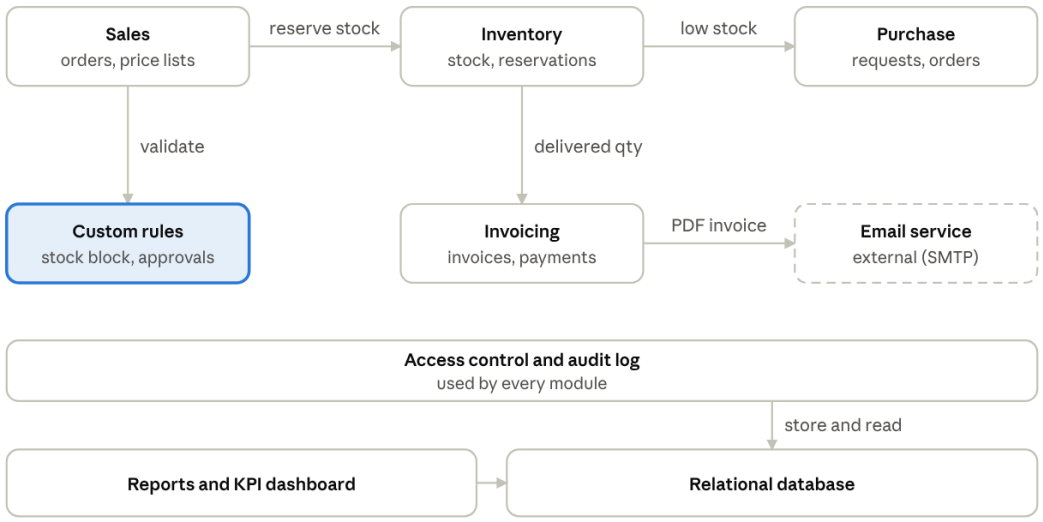


activity diagram · order confirmation checks

A.4 Component Diagram

The components below are the standard Odoo modules plus the custom rules and shared services this project adds. Arrows show which component calls which.

Four business modules depend on each other and on shared services



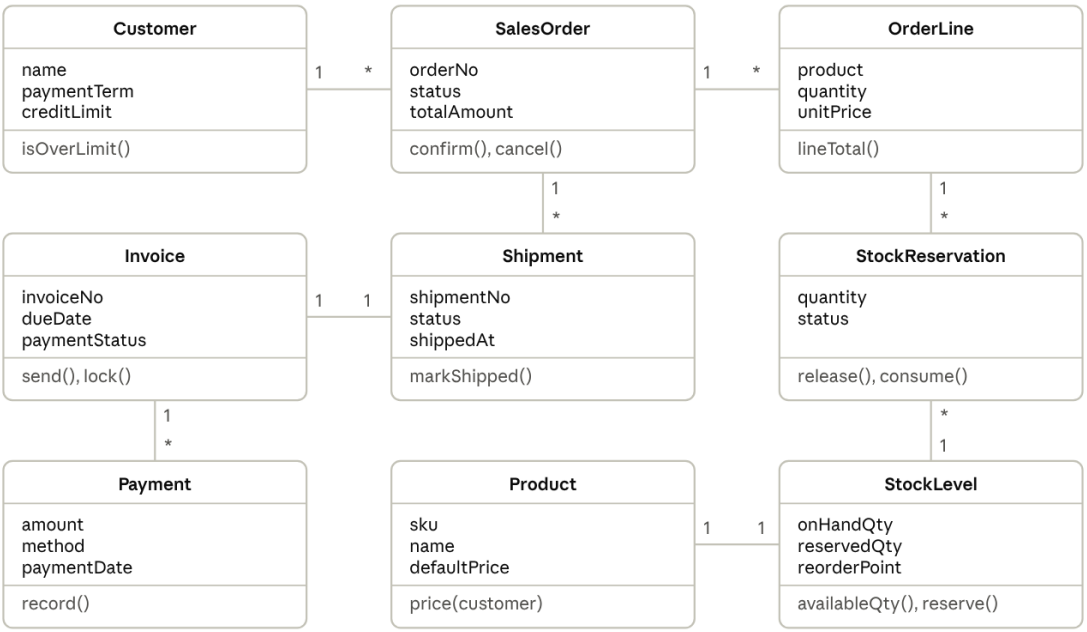
Dashed box: external service. Highlighted: custom rules added to standard Odoo.

component diagram · modules and shared services

A.5 Class Diagram (Core Domain)

The classes correspond to the main tables in the data model. Attribute lists are shortened to the fields that matter for the behaviour described in this document. Multiplicity: 1 means one, * means many.

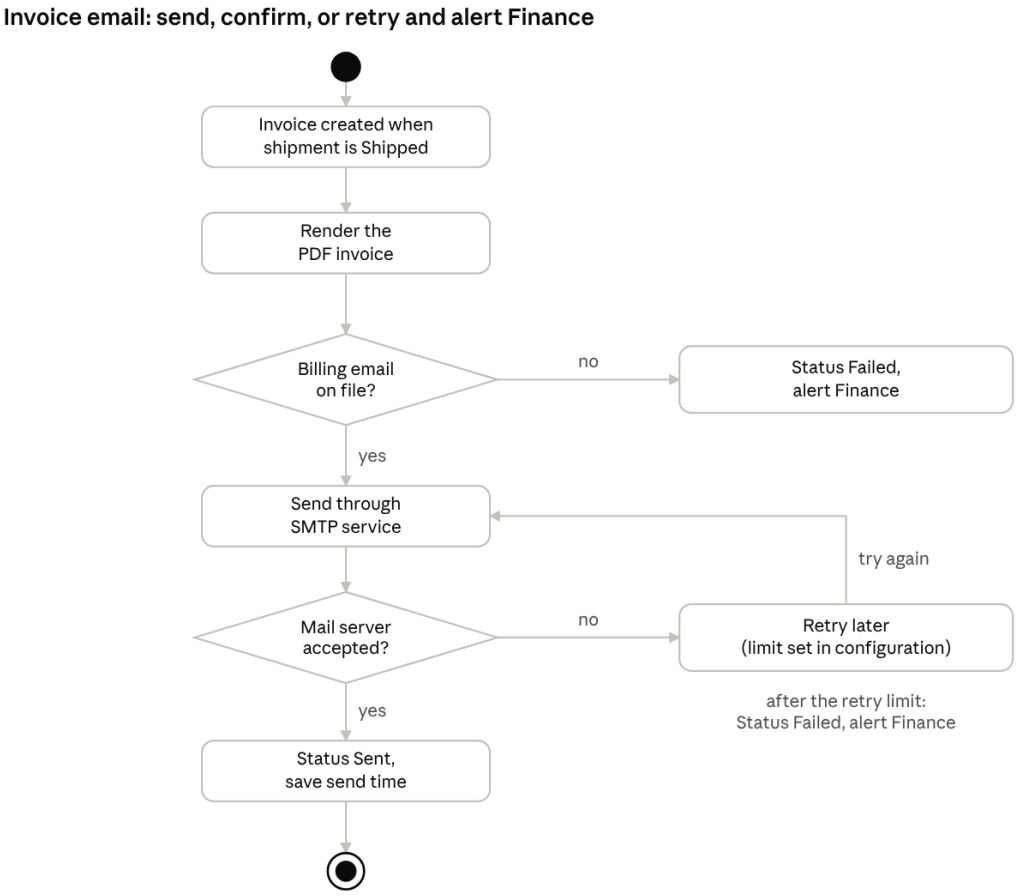
Nine core classes: an order has lines, shipments, and invoices



class diagram · nine core classes

A.6 Integration Flowchart: Invoice Email

The only external integration in phase 1 is the outgoing email service. There is no payment gateway and no WhatsApp integration. The retry limit is an open decision (see OI-05 and the email configuration).



integration flowchart · invoice email through SMTP